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Conformation-Mediated Doping in P3HT:F4TCNQ Dimers from Density Functional Theory

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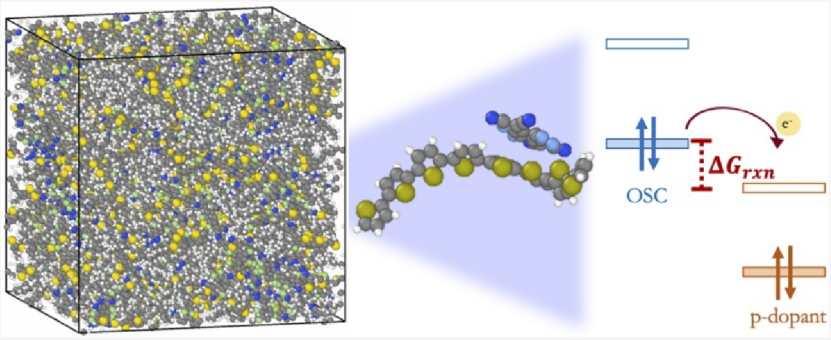

ABSTRACT: Molecular doping of charge carriers in organic semiconductors is a complex process influenced by both single-molecule energetics and multiscale morphology. Here, we employ atomistic molecular dynamics simulations and density functional theory (DFT) calculations of disordered P3HT:F4TCNQ morphologies to understand how conformational disorder influences doping efficiency. We find that conformational variations in P3HT:F4TCNQ dimers can modulate the amount of ground-state charge transfer between P3HT and F4TCNQ by more than 0.5 C. The amount of charge transfer in P3HT:F4TCNQ dimers exhibits a linear correlation with the difference between the ionization potential (IP) of P3HT and the electron affinity (EA) of F4TCNQ, when using geometries extracted from DFT-optimized dimer aggregates. We find that most variation in IP − |EA| that governs the amount of ground-state charge transfer is due to conformational variations of P3HT's IP (∼0.35 eV) compared with F4TCNQ's EA (∼0.15 eV). Finally, we show that the free energy of charge generation (ΔG_{rxn}) of the dimer, when treated as the sum of IP − |EA| and electrostatic energies derived from atomic partial charges, correlates linearly with the amount of charge transfer in the dimer. These findings shed light on the energetics of the molecular doping process, justifying simple approximations employed in recent reactive Monte Carlo methods for molecular doping and providing an avenue for systematic DFT parametrization of multiscale doping methods.

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■ INTRODUCTION

Organic semiconductors (OSCs) have garnered promise for a variety of commercial applications, including thin-film and flexible photovoltaics, light emitting diodes, and field-effect transistors.[1] While chemical doping plays a critical role in enhancing the performance of both inorganic and organic semiconductors, the doping mechanism in OSCs presents unique challenges due to structural and energetic disorder.[2,3] Unlike inorganic semiconductors with high dielectric constants and well-defined energy bands, OSCs have low dielectric constants and exhibit thermally driven charge transport.[4,5] Unlike inorganic semiconductors, which require only trace levels of dopants (∼10^{-5}%), OSCs require orders of magnitude higher concentrations (∼1%) to achieve comparable conductivities. This high loading of a molecular additive can severely alter the film morphology, creating conditions that reduce doping efficiency and hinder charge transport.[4,6−8]

Molecular doping in OSCs through charge transfer involves multiple electronic species, which confounds rational design. Two common charge transfer species involved in molecular doping are the integer charge transfer (ICT) state and the charge transfer complex (CTC). ICT states form through electron transfer between a donor−acceptor pair, producing a charge-separated state that can contribute to conductivity, although only a small fraction of these states (around 5%) dissociate into free carriers due to strong Coulomb interactions.[9] CTCs involve hybridization of the frontier

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molecular orbitals of the OSC and dopant, resulting in a neutral complex that requires thermal excitation for charge conductivity. Efforts have begun to exploit the use of various functional groups on the dopant to tune the ratio of ICTs to CTCs; notably, branched side chains force 2,3,5,6-tetrafluoro-7,7,8,8-tetracyanoquinodimethane (F4TCNQ, a common molecular dopant for OSCs) to reside between the π-faces of crystallites, and the alternating backbone energetics of OSC copolymer motifs can both induce CTC formation.[10,11] The doping mechanism itself depends on polymer microstructure and dopant chemistry; regioregular P3HT exhibits more delocalized polarons and consistent optical signatures across dopants, contrasting with regiorandom variants where charge-transfer complexes and localized polarons dominate.[12] Thus, tuning the populations of the different electronic states in molecular doping is of key importance to engineering molecularly doped OSC films,[1,13−18] but predicting the amount of ground-state charge transfer between an OSC and a dopant remains challenging.[19−25]

The most common design rule for predicting doping efficiency across a broad chemical space considers the amount of ground-state charge transfer to be proportional to the energetic offset between the single-molecule ionization potential (IP) of the OSC and the single-molecule electron affinity (EA) of the dopant (for p-type dopants), with the orbital overlap mediating CTC state formation. Experimental works focused on molecular engineering have led to significant chemical diversity in dopants with which to test this hypothesis, with moieties varying in size, charge, composition, atom type, and ring structures. Specific manifestations of these chemistries include TCNQ-derivatives, fullerenes, Lewis acids and bases, cationic dyes, organometlalics, radialenes, DMBI-derivatives, amines, polymers, self-dopants, and boron-containing materials.[4,6,26,27] While the considerations of single molecule energetics form a starting point for rational design, single-molecule energetics cannot capture the full complexity of the molecular doping process, and numerous works have discussed the critical role of multimolecule effects (notably Coulombic interactions in charge transfer states and environmental polarization).[5,21,28−30]

In considering rational OSC-dopant design, we were motivated to quantitatively assess the utility of the single-molecule energetic design strategy using literature data. As doping efficiency is assumed to be governed by the free energy of charge generation, typically approximated as $\Delta G_{rxn} = \mathrm{IP}_{\mathrm{OSC}} - \mathrm{EA}_{\mathrm{dopant}}$, we conducted a literature survey (see Table S1) of the reported experimental ΔG_{rxn} across 70 pairs of molecularly doped OSCs extracted from the literature (Figure 1). The majority of these energy level estimations are performed using cyclic voltammetry (CV) measurements of the individual OSC and dopant molecules. While doping efficiencies are not explicitly reported in all 70 works, we sought to examine if the effective dopants in the literature clustered around a particular range of ΔG_{rxn}, motivating the efficacy of single-molecule energetic considerations. Surprisingly, Figure 1 shows that the reported ΔG_{rxn} values in the literature estimated from single-molecule energy measurements span nearly 6 eV. Of similar importance is that the most common ΔG_{rxn} value found in the literature survey of molecular dopant-OSC combinations is slightly above zero, with some dopants reported with ΔG_{rxn} as high as +3 eV. For such large, positive ΔG_{rxn} values reported, one must consider the validity of these simple estimations of ΔG_{rxn} in experimental design.

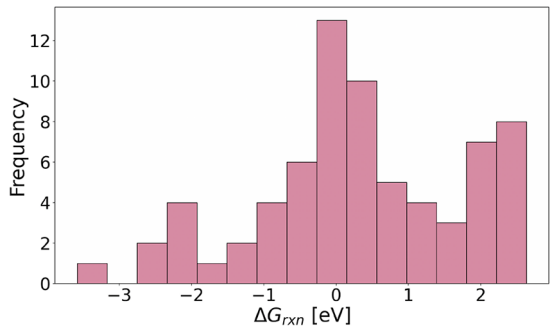

Figure 1. ΔG_{rxn} estimated for 70 OSC-dopant pairs extracted from the experimental literature.

Previous computational works have carefully elucidated many of the critical details of single molecule energy considerations for molecular doping.[31] Foundational work by Bredas leveraged density functional theory (DFT) and population analysis to show a correlation between the amount of ground-state charge transfer in an OSC-dopant dimer and the IP and |EA| of the OSC and dopant separately.[32] However, this linear relationship between the amount of ground-state charge transfer and IP − |EA| only emerged when using the single molecule geometries extracted from the DFT-optimized dimer geometry; the single molecule energetics of the OSC and dopant in isolation from each other did not correlate strongly with the amount of ground-state charge transfer in the dimer. A range of ΔG_{rxn} similar to that in Figure 1 was only observed by Zhu et al. when scaling the DFT-calculated IP and |EA| by a 1.6 eV shift to qualitatively account for polarization effects, though this was stated to be a coarse approximation of the true multimolecule environment. Importantly, this work showed that single-molecule energetics must be carefully evaluated when employed as an experimental design rule, and considerations of dimer aggregation, charge transfer, and environmental polarization play a critical role in the ultimate charge transfer or doping efficiency.[17,28,30,32−34] In some cases, the |EA| of a dopant impurity within an OSC local environment can be up to 1 eV lower than that of the pure dopant crystal.[35] It is also important to note that similar thematic considerations played a critical role in understanding charge generation efficiency in the organic solar cell community a decade ago.[28,31]

A simple way of improving upon single-molecule energetics estimates of ΔG_{rxn} involves explicitly introducing the Coulomb interaction between electron and hole. For a model system of an OSC and dopant dimer, and a dielectric constant ranging between 1 and 4, the Coulomb interaction between electron and hole can be ∼1 eV. It is important to note that foundational work in OSC-dopant design has leveraged bulky dopants to directly modulate this Coulomb binding energy to increase mobilities. With regard to the experimental ΔG_{rxn} values in Figure 1, a subtraction of ∼1 eV renders ΔG_{rxn} across the literature significantly more exothermic. Exploiting this simple physics governing doping was recently leveraged in a reactive Monte Carlo model to understand the morphology dependence of molecular doping, as well as the doping dependence of the morphology, for a simple Lennard-Jones fluid representation of P3HT:F4TCNQ.[36] Incorporating further complexity of condensed phase polarization with first-principles effects likely shifts the IP - |EA| toward more negative values, thereby further stabilizing the ground-state

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charge transfer and making the process more exothermic. These results suggest that single-molecule energy estimations only partially correlate with doping efficiency, as they neglect many-body effects from the molecular environment and conformational variations.

A substantial body of emerging theoretical work has highlighted that molecular doping is governed by condensed-phase physics rather than the energetics of an isolated donor−acceptor dimer.[37,38] Among these notable works, Cocchi et al. demonstrated through many-body GW-BSE calculations on oligothiophene:F4TCNQ stacks that donor length modulates the electronic structure of CTCs, renormalizing charge-transfer (CT) states by several hundred meV.[23] Even with identical energetic offsets, the local donor−acceptor geometry (e.g., pi-stacked vs side-chain interactions) critically determines the amount of ground-state charge transfer, highlighting the importance of configurational disorder.[39] In fact, systematic sampling of a broad range of geometries in PCPDT-BT copolymers showed that the presence of ICT states hinges on the local polymer conformation that surrounds each dopant.[40] Further work applying DFT to clusters has confirmed that two-dopant/two-polymer complexes reduce Coulomb binding and open "double-ICT" channels.[41] Hybrid QM/MM embeddings have since mapped complete ICT binding-energy distributions in n-doped hosts and shown how dielectric disorder and packing heterogeneity cap doping efficiency.[42] A multiscale QuantumPatch to kMC workflow then connected those distributions to macroscopic conductivity, attributing order-of-magnitude gains to short-range "overscreening" driven by the dopant quadrupole tensor.[33] Independent microelectrostatic studies corroborate this dual role of quadrupoles, showing that they both ease dopant ionization and weaken carrier binding,[43] while hybrid quantum-classical calculations on semicrystalline PBTTT locate the most effective acceptors in alkyl-rich regions where environmental screening is strongest.[16] Fully atomistic molecular dynamics (MD) free-energy profiles for F4TCNQ-doped P3HT went further, revealing that the entropic penalty for dopant aggregation competes with the enthalpic driving force for ICT and explains the sublinear conductivity-versus-loading curves often seen experimentally.[21] While these studies have added critical understanding to the literature, these works have predominantly leveraged idealized dimer configurations as opposed to those extracted from realistic molecular simulations of packing. Addressing this gap is essential to capture the nuances of molecular doping in OSCs.

The aim of this work is to investigate the role that conformational variations in OSC-dopant dimers play in dictating charge transfer by examining highly disordered OSC-dopant morphologies that sample an intentionally diverse conformational space using atomistic MD simulations of P3HT:F4TCNQ. First, we describe the atomistic MD methodologies to generate OSC-dopant dimer configurations and the DFT-based methods for computing their electronic structures. We then quantify the distribution of the amount of ground-state charge transfer over these dimers using explicit geometry optimization with DFT, which allows polaronic relaxation and enables meaningful charge transfer. We then dissect the energetics associated with this charge transfer process from the perspective of single-molecule energetics and corrections based on atomic point charge-based Coulombic interactions. A linear correlation between the amount of charge transfer and ΔG_{rxn} estimated using IP − |EA| and a point charge-based electrostatics term motivates the use of simple approximations for understanding reactivity of molecularly doped semiconductors in different morphologies.

■ METHODS

Atomistic Molecular Dynamics Simulations. Diverse P3HT:F4TCNQ dimer configurations were extracted from MD simulations of a disordered P3HT:F4TCNQ morphology. To simulate the P3HT:F4TCNQ morphology, we randomly arranged 100 8-mer P3HT oligomers and 120 F4TCNQ molecules within a cubic box (50 Å) using PACKMOL,[44] creating five initial configurations with different velocity initializations and random packings (Figure 2). While this concentration of F4TCNQ within P3HT is unrealistic, it enables the robust generation of dimer configurations that would be highly inefficient if simulations were conducted at experimental doping concentrations of e.g., ∼1%. We do not anticipate the distribution of local dimer configurations to be strongly affected by the artificially high dopant concentration, as these simulations occur in an amorphous morphology over time scales too short for larger scale ordering or crystallization. Each configuration was initially relaxed at 400 K and 100 bar for 0.2 ns in the NPT ensemble, controlled by a Nosé-Hoover thermostat and barostat to reduce the density. This was followed by a 40 ns simulation at 400 K and 1 bar, after which the system was annealed with a 40 ns run where the temperature was linearly reduced from 400 to 300 K, maintaining pressure at 1 bar, to achieve a realistic density ($\sim 1.1\ \frac{g}{cm^3}$). Subsequently, a final 20 ns equilibration was conducted at 300 K and 1 bar. After the thermal annealing process, we performed an additional 20 ns simulation at 300 K and 1 bar in the NVT ensemble for data collection. The thermostat and barostat coupling constants were set to 1.0 ps for both NPT and NVT ensembles.

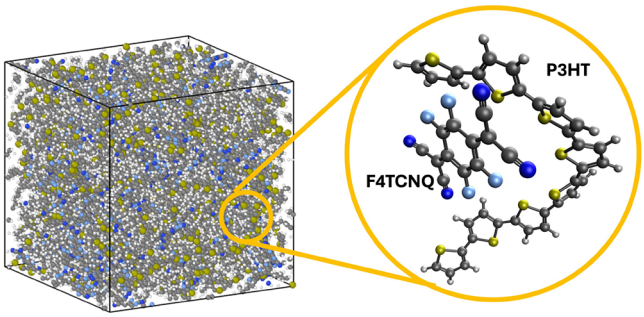

Figure 2. Snapshot from an MD simulation showing an F4TCNQ−P3HT dimer. Carbon, sulfur, nitrogen, and fluorine atoms are depicted in gray, yellow, dark blue, and sky blue, respectively. For clarity, the alkyl side chains of P3HT have been omitted.

MD simulations used the velocity Verlet algorithm with a time step of 1 fs and periodic boundary conditions in all three dimensions. For long-range electrostatic interactions, we employed the particle−particle particle-mesh (PPPM) method with a k-space accuracy of 0.0001, applying a real-space cutoff of 12 Å. The same cutoff distance was used for Lennard-Jones interactions, with geometric mixing rules for interactions between different atom types. For force field parameters, an 8-mer of P3HT was modeled using the OPLS-AA force field as proposed in Ref. 45, which is optimized to replicate the crystal structures of alkylthiophene-based oligomers and polymers.

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For F4TCNQ, Lennard-Jones parameters and harmonic stretching, angle bending, and torsional parameters were derived from the standard OPLS-AA force field.[46] Atomic point charges for F4TCNQ were calculated by fitting the electrostatic potential to the optimized geometry at the ωB97XD/cc-pVDZ level of theory, with this geometry also defining the equilibrium bond lengths and angles. All MD simulations were performed using the LAMMPS software package;[47] the geometry optimization of F4TCNQ was conducted using ORCA.[48]

Density Functional Theory Calculations. To determine the amount of ground-state charge transfer in P3HT:F4TCNQ dimers, DFT calculations were performed on 250 MD-derived dimer configurations of P3HT:F4TCNQ. For each configuration extracted from MD, a geometry optimization (ωB97XD/6-31+G* using Gaussian 16[49]) was performed. This level of theory was validated against the smaller cc-pVDZ basis set, where the difference in computed electrostatic potential energies associated in charge transfer states in this work was found to be only ∼0.01 eV. The ωB97XD functional was chosen due to its inclusion of empirical dispersion corrections, which are critical for accurately modeling non-covalent interactions in donor−acceptor systems.[32] A polarizable continuum model using the integral equation formalism variant (IEFPCM)[50,51] with a dielectric constant of 4.3 was used to approximate the bulk OSC environment around the dimer. The DFT geometry optimization was observed to be critically important to observe the difference between the net charge on P3HT and F4TCNQ (denoted as ΔQ) larger than 0.1 C (see Figure S10). This need for DFT-optimized geometries to observe large ΔQ values could be due to two reasons. First, the force-field's minimum energy geometries do not exhibit sufficient overlap with those derived from DFT, and thus charge is only transfered once this geometric reorganization can take place. Second, there is the need to allow for polaronic contributions to ground state charge transfer (i.e., the value of ΔQ is stabilized by enabling vibrational relaxation within the DFT geometry optimization). Provided the parametrization details of the oligothiophene force-field, we are inclined to believe that polaronic effects are the dominant contributor to observing large ΔQ in DFT-optimized geometries.The difference between MD-extracted geometries and DFT-optimized geometries is visualized and quantified in Figure S2.

The individual P3HT and F4TCNQ molecules from the optimized dimer were then extracted, and single-point DFT calculations were performed on each individual molecule in its neutral and charged states (0 and +1 for P3HT, 0 and −1 for F4TCNQ) to determine the IP and |EA| of P3HT and F4TCNQ, respectively. Natural Population Analysis (NPA) was then conducted on the neutral DFT-optimized dimer geometries to determine the charge localization on each molecule. NPA uses an orthonormal set of localized functions resembling atomic orbitals (natural atomic orbitals) to partition the molecular electron density and assign chemically intuitive atomic charges, and is more reliable for aggregates than population analysis techniques based on basis function partitioning or electrostatic potential fitting approaches. For further interpretation beyond the single point level of theory, the energy of the system was further decomposed using sobEDA[52] and the correlation of each component to the observed amount of charge transfer can be found in the SI.

RESULTS AND DISCUSSION

DFT-based population analysis on the P3HT:F4TCNQ dimer configurations is used to understand how conformational variations in the dimer aggregate affect the amount of ground state charge transfer. For each extracted dimer geometry that has been optimized in the neutral state, we sum the NPA-determined atomic partial charges of the P3HT and F4TCNQ, respectively, to extract a net charge per molecule. Taking the difference between the net charge on P3HT and F4TCNQ, denoted ΔQ, serves as a quantitative characterization of the amount of ground state charge transfer within the dimer. Despite the implicit solvent environment and fixed material pair, ΔQ varies significantly across the 250 configurations, ranging from 0.0 to 0.55 C (Figure 3). This result is consistent with conformational variations discussed in prior experimental works that observed broad distributions in IP of OSCs, which were posited to be the result of substantial conformational disorder.[8,53−55] The significant variation in ΔQ, even within a fixed pair of OSC and dopant chemistry, highlights the role of multimolecule effects beyond single molecule energetics in dictating the amount of ground state charge transfer, and thus the doping efficiency.

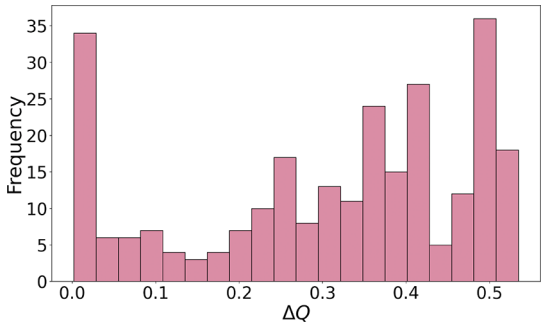

Figure 3. Histogram of the amount of ground-state charge transfer, ΔQ, observed across 250 MD-extracted, DFT-optimized dimer configurations, as measured by the difference in NPA-derived charges between P3HT and F4TCNQ.

We next investigated whether this purely conformationally induced variation in ΔQ correlated with the DFT-derived total energies of the optimized dimer configurations. If $\Delta Q > 0$ is energetically favorable, we expect to see larger values of ΔQ positively correlated with lower total energies of the system. Prior work has demonstrated that structural order can stabilize fractional charge transfer via strong orbital hybridization and dense molecular packing, indicating that local geometry not only affects the amount of charge transfer but can also energetically favor intermediate (noninteger) CT states over full electron transfer.[22] In Figure 4 it is observed that ΔQ positively correlates with a decrease in the total DFT-derived energy of the dimer configurations. Interestingly, the value of ΔQ observed across all P3HT:F4TCNQ dimer conformations studied reaches a maximum at approximately 0.55 C, but this does not represent a local minimum of the total energy of the dimer. This suggests that even larger ΔQ could potentially lower the system's total energy, though it is unclear how to access these lower energy dimer conformational states. Such an analysis would also be ideally suited to a method such as constrained DFT,[56] where the energetics of partial charge transfer could be tracked, but this is beyond the scope of the present study.

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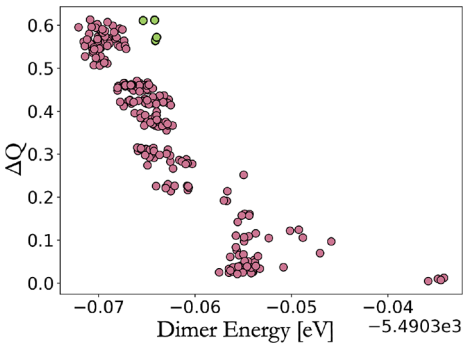

Figure 4. Scatter plot of the amount of ground-state charge transfer, ΔQ, in DFT-optimized P3HT:F4TCNQ dimer geometries versus the total DFT-derived energy of each dimer geometry, shifted relative to the minimum energy dimer configuration in the data set (R^2 = 0.8084). Points in green denote outliers where the value of ΔQ is maximized, yet the total dimer energy does not reach a local minimum.

Motivated by this linear relationship between ΔQ and the DFT total energy of the dimer, we examined if the IP − |EA| contribution to the total energy also exhibited a linear relationship with ΔQ, as mentioned in the previous work of Zhu et al.[32] Specifcally, Zhu et al. have demonstrated that, across various OSC and dopant chemistries using idealized geometries, a linear relationship exists between IP − |EA| and ΔQ when using the DFT-optimized dimer geometries, corroborating experimental observations of charge transfer to F4TCNQ from thiophene (co)polymer matrices being determined by the IP derived from a local thiophene segment.[57] This correlation is typically attributed to the alignment of frontier orbital energetics, which governs the ease of charge separation in donor−acceptor interfaces. Based on this framework, we expected to observe a similar trend in our system as we conformationally vary the IP within a single material pairing. To examine this, the single molecule configurations of P3HT and F4TCNQ were extracted from the optimized dimers, and single point calculations of the neutral and ionized states of the isolated individual molecules were performed to assess the diabatic IP and EA. In Figure 5, We reproduce a roughly linear relationship between IP − |EA| and ΔQ (R^2 = 0.89). Deviations from linearity are observed for larger IP − |EA|, indicating that other factors such as molecular packing, intermolecular interactions, and the formation of

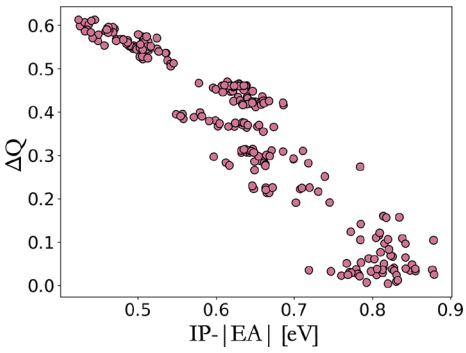

Figure 5. Correlation between the ionization potential-electron affinity (IP − |EA|) energy gap and the amount of ground-state charge transfer, ΔQ, in the P3HT:F4TCNQ dimer system (R^2 = 0.8906).

charge transfer complexes influence ΔQ as the IP − |EA| gap increases.

With the substantial variations in IP − |EA| across a diverse configurational space of P3HT:F4TCNQ dimer aggregates, we next evaluated to what degree the P3HT and F4TCNQ moieties were responsible for the variations in ΔG_{rxn}. Figure 6

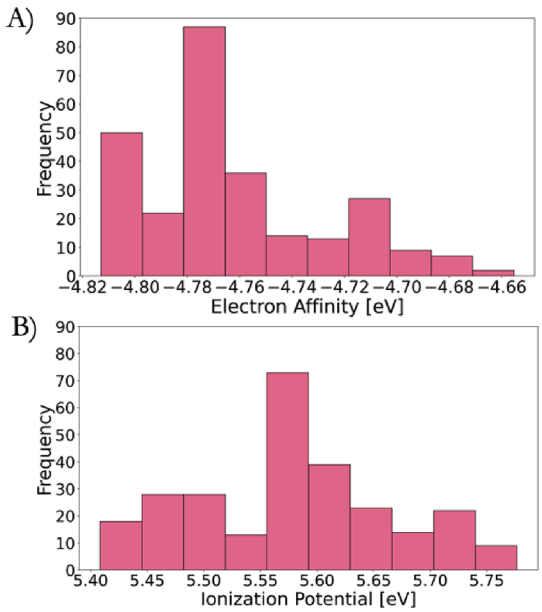

Figure 6. Statistics of conformationally induced variations in the (A) electronic affinity of F4TCNQ and (B) ionization potential of P3HT for geometries extracted from DFT-optimized dimer geometries.

shows that substantially more variation in this energetic quantity can be ascribed to the conformational variation of P3HT (∼0.35 eV) relative to F4TCNQ (∼0.15 eV). This is likely attributable to the greater conformational flexibility of P3HT through its interring dihedrals relative to the rigid and planar structure of F4TCNQ. As the IP of P3HT is lowered by increasing planarity of the π-electron system, it is natural to anticipate that more planar P3HT configurations lead to enhanced charge transfer. This hypothesis is partially supported (Figures S1 and S2), where it is observed that the amount of charge transfer in the dimer weakly correlates with a decrease average value of the interring dihedral angles in P3HT, as well as the separation distance between the P3HT and F4TCNQ centers of mass. However, it is important to note that these correlations are weak, and a variety of more complex geometric considerations are likely participating (Figure S3). In an attempt to capture these more complex interactions, we leveraged principal components analysis and t-stochastic neighborhood embedding on the atomic distance matrix of the dimer configurations, but witnessed little clustering of the dimer geometries that correlated with the amount of charge transfer observed in the system (Figures S4 and S5).

As isolated single-molecule energies are insufficient to describe the complex dependence of charge transfer on the dimer's electronic structure, we sought to assess whether atomic point-charge based intermolecular electrostatic energies, when added to single-molecule IP − |EA| quantity, would render the calculated ΔG_{rxn} exothermic. For reference, note that in Figure 5 the scale of the energy axis ranges between 0.6 and 1.1 eV, being strongly endothermic. This energetic scale is

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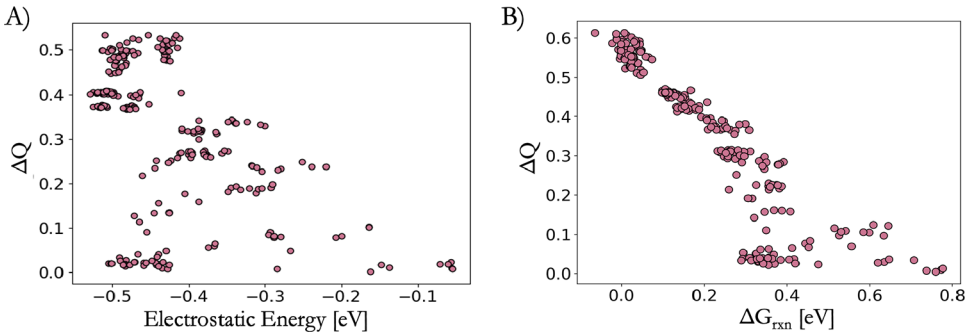

Figure 7. Amount of ground-state charge transfer, ΔQ, in the dimer geometry plotted against (A) the intermolecular electrostatic energy between P3HT and F4TCNQ evaluated using NPA-derived atomic partial charges (R^2 = 0.3305) and (B) ΔG_{rxn}, where ΔG_{rxn} is determined by the difference between the ionization potential and electron affinity summed with an atomic point charge-based intermolecular electrostatic interaction (R^2 = 0.8345).

qualitatively consistent with the IP − |EA| estimate of ΔG_{rxn} from Figure 1. In Figure 7A), we plot the correlation between ΔQ in the dimers and the total intermolecular electrostatic energy obtained by evaluating the NPA-derived partial charge interactions in the DFT-optimized dimer configuration between P3HT and F4TCNQ (intramolecular P3HT and F4TCNQ Coulomb interactions are omitted). Intermolecular electrostatic energies in these systems range from −0.5 to 0.0 eV, consistent with expectations based on a simple estimate of the Coulomb energy between two point charges separated by ∼1 nm. It is generally observed that large negative electrostatic interaction energies are correlated with a larger ΔQ. However, we note that for systems with $\Delta Q \approx 0$, the magnitude of the electrostatic energy was much more varied. It is also important to recognize that subtracting the atomic point charge based intermolecular electrostatic energy from the total DFT energy could enable fitting of the ΔG_{rxn} dependence of charge transfer, which is directly utilizable within reactive Monte Carlo (RMC).

To further dissect the electrostatic potential energy of interaction within the dimers, we superimposed two atomic point charge distributions onto the dimer geometries to assess how much the single-molecule atomic point charge distributions in the neutral and doped states resemble those from the dimer optimized state. Beyond fundamental understanding, this task is also relevant for the superposition of classical electrostatic corrections to ΔG_{rxn} being developed for RMC methods,[36] and is also important provided recent literature on electrostatic screening efforts in Coulombic binding energies.[33] For all DFT-optimized neutral dimer configurations, we replaced the P3HT:F4TCNQ atomic point charges with those extracted from isolated single-molecule DFT calculations in the neutral and cation/anion states. We then plotted the atomic partial charge derived electrostatic energies for each dimer configuration against the isolated doped and undoped state atomic partial charge derived electrostatic energies in those same dimer configurations (Figures S6 and S8). Interestingly, electrostatic energies derived from the dimer DFT calculations exhibit strong linear correlation with those energies derived using the undoped partial charges of P3HT and F4TCNQ in isolation, even for doped states up to 0.55 C. Contrastingly, the electrostatic energies derived from the dimer DFT atomic partial charges exhibit a linear correlation with a different slope at large values of ΔQ, but no correlation with those exhibiting $\Delta Q \approx 0$. We further explored these relationships by calculating the HOMO−LUMO intermolecular electronic coupling via the dimer projection method (Figure S9), but these results showed no strong correlation between ΔQ and the magnitude of the intermolecular electronic coupling.

We next assessed how much the atomic point charge-derived electrostatic interaction energy in the doped state of the P3HT:F4TCNQ dimer would shift the calculated ΔG_{rxn} from its previously highly endothermic values. In Figure 7B) we sum the electrostatic energies from Figure 7A) with the energies from Figure 5 to correlate the corrected ΔG_{rxn}. It is notable that now the calculated ΔG_{rxn} spans 0.1−1.0 eV, which is much closer to a thermally accessible free-energy for charge transfer than via the naive single-molecule energy considerations, but is still significantly endothermic.

The role of polaronic effects in P3HT:F4TCNQ doping is notable, as we find that no appreciable amount of charge transfer (ΔQ < 0.1C) occurs for the raw MD-extracted P3HT:F4TCNQ dimers (see Figure S10), even in the presence of a continuum polarizable solvent, yet allowing the pair to relax at the DFT level immediately stabilizes large ΔQ. This is evidence that the electronic reorganization energy (≈ 200−300 meV in thiophene systems) must be accurately treated before doping can proceed, in agreement with the work of Li et al.[58] Spectroscopic fingerprints echo this picture: mid-IR and IR-active-vibration bands that grow in doped P3HT track the progressive self-localization of the hole and its accompanying backbone distortion,[59] while near-IR transient absorption captures a subpicosecond growth of the polaron band followed by slower structural relaxation, underscoring the multitime scale nature of polaron formation.[60] Quantitative models must therefore pair explicit dimer-level codesign with polaronic reorganization energies, for example through constrained-DFT or many-body perturbation schemes that embed soft-phonon modes, to reach predictive discovery of dopant−semiconductor combinations.

Designing molecularly doped organic semiconductors demands treating the OSC host and dopant as a single supramolecular entity whose electronic structure is fixed by the dimer geometry, not by isolated monomer energetics. Early DFT work on F4TCNQ:oligothiophene pairs showed that ΔQ tracks the IP − |EA| gap only when both quantities are taken from the dimer-relaxed geometries, revealing how orbital hybridization and geometry-dependent polarization reshape the thermodynamics of doping.[32] Follow-up spectroscopic studies on donor−acceptor copolymers demonstrated that sliding a dopant just a few Angstroms along the backbone

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moves the complex between fractional and integer charge-transfer states, highlighting the decisive role of intermolecular orientation.[10,40] In semicrystalline P3HT, polymorphic packing yields two distinct charge-transfer phases with conductivities that differ by 2 orders of magnitude — despite identical monomer energy levels — again underscoring the structural origin of doping efficiency. Moreover, recent multiscale simulations have shown that short-range overscreening driven by the dopant anion's quadrupole moment and dimer orientation can tune the Coulomb binding energy by hundreds of meV, offering a geometric lever for boosting carrier densities.[33,43] When combined with the conformational dependence of the doping energetics in this work, these results make a compelling case that OSC:dopant codesign must explicitly target favorable dimer geometries — integrating steric, electrostatic and electrostatic considerations — to move beyond the heuristics of isolated single-molecule energetics and toward predictive discovery of codesigned OSC-dopant pairs.

The current survey of P3HT:F4TCNQ dimers demonstrated that once each dimer is allowed to undergo its local polaronic relaxation, the thermodynamic driving force for doping can be written as linearly proportional to the single-molecule |IP − EA| gap plus a classical Coulomb term utilizing atomic point charges: this simple expression reproduces the full quantum-chemical ΔG_{rxn} and tracks ΔQ across the entire conformational ensemble. Because the Coulomb stabilization can exceeds 1 eV for realistic intermolecular separations in a low-dielectric ($\varepsilon \approx 1 - 4$) organic environment, the fully relaxed ICT state is always lower in energy than either the neutral dimer or fractional charge-transfer complexes. These findings justify the key idealizations built into the reactive Monte Carlo (RMC) model:[36] each OSC or dopant is coarse-grained to an isotropic Lennard-Jones bead that can carry 0 or ±e, and the reaction free energy is taken as ΔG_{rxn} = IP − |EA| with the Coulomb attraction handled by an Ewald-summed point-charge interaction.[36] The large electrostatic stabilization quantified in the present work explains why the RMC simulations predict near-unity doping at $\varepsilon = 3$ even when the underlying single molecule IP − |EA| is positive, and why the simulated doping efficiency becomes sensitive to ΔG_{rxn} only after the dielectric is raised to unphysically large values, exactly mirroring the dimer-level energetics. In essence, the present detailed DFT study shows that once local polaronic effects are absorbed into an effective ΔG_{rxn} parameter, a coarse-grained model retaining only (i) single-molecule redox energies, (ii) dielectric-screened Coulomb interactions, and (iii) steric exclusion is capable of capturing much of the dominant thermodynamic drivers of molecular doping, thereby validating the simplifying assumptions of the RMC framework. Moving forward, replacing the simple monopole-based electrostatics term of the RMC model with a classical electrostatics treatment using atomic partial charges will likely be sufficient to capture the physics of molecular doping when combined with the appropriate single molecule energetics terms.

CONCLUSION

In summary, our atomistic MD and DFT simulations on disordered P3HT:F4TCNQ morphologies reveal that conformational variations in donor−acceptor dimers significantly modulate the amount of ground-state charge transfer, ΔQ, yielding a broad distribution up to 0.55 C. We demonstrate a linear correlation between the transfer charge and the difference in the IP of P3HT and EA of F4TCNQ, with P3HT's conformational flexibility primarily dictating the IP − |EA| variation. Crucially, the thermodynamic driving force (ΔG_{rxn}) for doping, when augmented by a classical electrostatic term derived from atomic point charges, correlates linearly with ΔQ, thereby justifying the simplifying assumptions employed in recent RMC models for molecular doping. This work underscores the critical importance of explicitly considering dimer aggregates and their conformational landscape in the codesign of organic semiconductor-dopant pairs, moving beyond simplistic single-molecule energetic considerations for predicting doping efficiency.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpca.5c05040.

Experimental OSC-dopant data set creation, correlations between charge transfer and dimer geometry, visualization of dimer conformations, dimensionality reduction, electrostatic energy evaluations for fully doped and undoped states, HOMO−LUMO electronic couplings for dimer aggregates, charge transfer in the unrelaxed dimer geometries (PDF)

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Notes

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